

A MACHINE LEARNING FRAMEWORK FOR BIOMETRIC AUTHENTICATION USING ELECTROCARDIOGRAM

*A Mini Project Report submitted to
JNTU Hyderabad in partial fulfilment
of the requirements for the award of the degree*

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

GHATTI VENKATA SIVANY

22RG1A0424

GANENA GAYATHRI

22RG1A0422

GOGOORI UMA DEVI

22RG1A0425

SABBARAPU YUKTHIKA

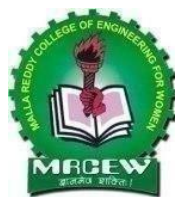
22RG1A0452

Under the Guidance of

Dr. SELVAMANI INDRAJITH

B.E., M.E., Ph.D.

Professor & HOD



**DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
MALLA REDDY COLLEGE OF ENGINEERING FOR WOMEN**

An UGC Autonomous Institution

Approved by AICTE New Delhi and Affiliated to JNTU

Maisammaguda, Medchal (Dist.), Hyderabad-500100, Telangana.

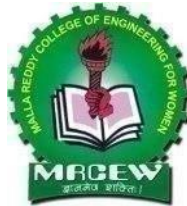
JUNE-2025

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
MALLA REDDY COLLEGE OF ENGINEERING FOR WOMEN

An UGC Autonomous Institution

*Approved by AICTE New Delhi and Affiliated to JNTU
Maisammaguda, Medchal (Dist.), Hyderabad-500100, Telangana.*

JUNE-2025



CERTIFICATE

This is to certify that the Mini Project entitled “**A MACHINE LEARNING FRAMEWORK FOR BIOMETRIC AUTHENTICATION USING ELECTROCARDIOGRAM**” has been submitted by **Ghatti Venkata Sivany (22RG1A0424)**, **Ganena Gayathri (22RG1A0422)**, **Gogoori Uma Devi (22RG1A0425)**, **Sabbarapu Yukthika (22RG1A0452)** in partial fulfilment of the requirements for the award of **BACHELOR OF TECHNOLOGY** in **ELECTRONICS & COMMUNICATION ENGINEERING**. This record of bonafide work was carried out by them under my guidance and supervision. *The result embodied in this mini project report has not been submitted to any other University or Institute for the award of any degree.*

Dr.Selvamani Indrajith

*Professor
Project Guide*

Dr.Selvamani Indrajith

Head of the department

External Examiner

ACKNOWLEDGEMENT

The Mini Project work carried out by our team in the Department of Electronics and Communication Engineering, Malla Reddy College of Engineering for Women, Hyderabad. ***This work is original and has not been submitted in part or full for any degree or diploma of any other university.***

We wish to acknowledge our sincere thanks to our project guide **Dr. Selvamani Indrajith**, Professor and Head of the Electronics & Communication Engineering Department for formulating the problem, analysis, guidance and his continuous supervision during the project work.

We acknowledge our sincere thanks to **Dr. Vaka Murali Mohan**, Director, MRCEW. **Dr. Kanaka Durga Returi**, Principal and Professor of Computer Science & Engineering Department. **Dr. Selvamani Indrajith**, Professor and Head of the Electronics & Communication Engineering Department, for their kind cooperation in making this Mini Project work a success.

We extend our gratitude to **Sri. Ch. Malla Reddy**, Founder Chairman and **Sri. Ch. Mahender Reddy**, Secretary, MRCEW and **Sri. P. Praveen Reddy**, Director, MRGI for their kind cooperation in providing the infrastructure for successful completion of our Mini Project work.

We convey our special thanks to the Entire Teaching faculty and non-teaching staff members of the Electronics & Communication Engineering Department for their support in making this project work a success.

GHATTI VENKATA SIVANY	(22RG1A0424)	_____
GANENA GAYATHRI	(22RG1A0422)	_____
GOGOORI UMA DEVI	(22RG1A0425)	_____
SABBARAPU YUKTHIKA	(22RG1A0452)	_____

INDEX

Chapter	Page No
ABSTRACT	iii
LIST OF FIGURES	iv
1. INTRODUCTION	1
2. LITERATURE SURVEY	3
3. EXISTING SYSTEM	5
4. PROPOSED SYSTEM	6
5. MODULE IMPLEMENTATION	8
5.1 DATA FLOW DIAGRAM	8
5.2 CLASS DIAGRAM	10
5.3 SEQUENCE DIAGRAM	11
5.4 USE CASE DIAGRAM	13
5.5 ACTIVITY DIAGRAM	15
6. SYSTEM REQUIREMENTS	17
6.1 HARDWARE REQUIREMENTS	17
6.2 SOFTWARE REQUIREMENTS	17
7. RESULT AND ANALYSIS	18
8. CONCLUSION & FUTURESCOPE	25
8.1 CONCLUSION	25
8.2 FUTURE SCOPE	25
9. APPENDIX	28
9.1 TECHNOLOGIES USED	28
9.2 Time Slicing and Machine Learning	31
9.3 Machine Learning Process for Training Data	33
9.4 Machine Learning for Classification	34
Approach	
9.5 Data Quality Measures	34
9.6 Mean Absolute Error Rate (MAER)	35
10. BIBILOGRAPH	37
11. YUKTI CERTIFECATE	39

ABSTRACT

This project introduces a framework for how to appropriately adapt and adjust machine learning (ML) techniques used to construct electrocardiogram (ECG)-based biometric authentication schemes. The proposed framework can help investigators and developers on ECG-based biometric authentication mechanisms define the boundaries of required datasets and get training data with good quality. To determine the boundaries of datasets, use case analysis is adopted. Based on various application scenarios on ECG-based authentication, three distinct use cases (or authentication categories) are developed. With more qualified training data given to corresponding machine learning schemes, the precision on ML-based ECG biometric authentication mechanisms are increased in consequence. The ECG time slicing technique with the R-peak anchoring is utilized in this framework to acquire ML training data with good quality. In the proposed framework four new measure metrics are introduced to evaluate the quality of the ML training and testing data. For developing ML-based ECG biometric authentication, the proposed framework can guide researchers to prepare the proper ML setups and the ML training datasets along with three identified user case scenarios. For researchers adopting ML techniques to design new schemes in other research domains, the proposed framework is still useful for generating the ML-based training and testing datasets with good quality and utilizing new measure metrics.

LIST OF FIGURES

FIG.NO	FIGURE NAME	PAGE NO
1.	Data flow diagram	8
2.	Class diagram	10
3.	Sequence diagram	11
4.	Use case diagram	13
5.	Activity diagram	15
6.	GUI of ECG based biometric authentication using Machine Learning	19
7.	ECG Dataset loaded in Authentication GUI	19
8.	Pre-processed ECG Dataset with Feature Count	20
9.	SVM Evaluation Result	21
10.	Evaluation of SVM and Decision Tree Models	21
11.	Evaluation report of SVM, Decision Tree and ANN	22
12.	Comparative Analysis of ML Models Using ECG Data	23
13.	User Interface for ECG Dataset processing and Model Training	23
14.	Performance Analysis of Machine Learning Models	24
15.	Time Sliced ECG data	33
16.	Training and Testing data	33

CHAPTER 1: INTRODUCTION

With the increasing reliance on internet-enabled systems, mobile applications, and digital platforms, the need for secure and reliable user authentication has become more important than ever. Users are frequently required to prove their identity to access services ranging from banking and healthcare to social media and e-commerce. Traditional authentication methods such as passwords, PINs, and security questions, while still widely used, have several drawbacks. These include vulnerability to theft, forgetting, and brute-force attacks. As cyber threats grow more sophisticated, these older methods are often no longer sufficient to ensure data and identity protection.

In response to these limitations, biometric authentication has emerged as a preferred solution for both security and usability. Biometric systems verify identity based on unique physical or behavioural traits, such as fingerprints, facial features, voice patterns, or iris scans. These methods are more secure because they are difficult to replicate or steal and offer the convenience of not needing to remember or enter credentials. As a result, biometric technologies have been rapidly adopted in personal devices, corporate environments, and government security systems. Their integration into smartphones and wearables has further increased user acceptance and accessibility.

Among the different biometric modalities, **electrocardiogram (ECG)-based authentication** is gaining significant attention for its unique benefits. Unlike static traits like fingerprints or facial structure, ECG is a dynamic signal that captures the electrical activity of the heart. This signal varies between individuals based on their anatomy and physiology, making it a strong candidate for unique identification. More importantly, ECG provides a measure of liveness — it proves that the user is alive and present at the time of authentication, which reduces the risk of spoofing attacks. This dynamic and time-dependent nature makes ECG particularly suitable for continuous authentication in secure environments.

To implement ECG-based authentication effectively, machine learning (ML) techniques are commonly used to build models that can recognize individuals based on their heart signals. ML algorithms learn from patterns in the ECG data, identifying subtle differences between users that are not easily detectable by traditional signal processing methods. Researchers have applied various supervised learning methods such as Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbours (KNN) to classify users with high accuracy. In recent years, deep learning models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks have shown even better performance due to their ability to automatically extract complex features from raw ECG signals.

Despite these advancements, several challenges remain in the path toward practical deployment of ECG-based biometric systems. ECG signals are sensitive to noise, body movement, electrode placement, and the user's emotional or physical state, all of which can affect the consistency of the data. Additionally, differences in hardware and sensors across devices make it difficult to ensure cross-device compatibility. Real-time use also demands lightweight and efficient models that can run on portable or wearable devices with limited computational power. To address these issues, ongoing research is focused on exploring more robust machine learning techniques, improving generalization across users and conditions, and developing privacy-preserving models that ensure user data remains secure. As these challenges are overcome, ECG-based authentication is expected to play a vital role in the next generation of secure and user-friendly identity verification systems.

CHAPTER 2: LITERATURE SURVEY

Zhang, Dzho, and X. Zeng Heart ID [1]

In 2017, Zhang, Dzho, and X. Zeng published a paper titled *Heart ID* in the journal *IEEE Access*. The study introduced a multiresolution convolutional neural network (CNN) designed for human identification using ECG (electrocardiogram) signals. This method uses heart data as a biometric to recognize individuals. It is especially useful for smart health applications where secure and accurate identification is important.

J.R. Pinto, J. S. Cardoso, and A. Lourenço [2]

The paper by J.R. Pinto, J.S. Cardoso, and A. Lourenço, published in *IEEE Access* in 2018, provides a comprehensive overview of ECG biometrics. It traces the evolution of the field, emphasizing the unique advantages of ECG signals for biometric recognition. The authors discuss key challenges such as intra-subject variability, noise, and acquisition constraints.

E.J. da Silva Luz, G.J.P. Moreira, L.S. Oliveira, W.R. Schwartz and D.Menotti [3]

E.J. da Silva Luz and colleagues published a study in *IEEE Transactions on Information Forensics and Security* in May 2018. The paper focuses on learning deep representations for heart biometrics using off-the-person ECG data. It emphasizes the use of deep learning techniques to extract robust features for identification.

H. Kim and S.Y. Chun [4]

H. Kim and S.Y. Chun presented a study in *IEEE Access* in 2019 on cancellable ECG biometrics. The work introduces a method combining compressive sensing with a generalized likelihood ratio test (GLRT). This approach enhances both security and privacy by allowing biometric templates to be revocable.

Y. Xin, L. Kong, Z. Liu, Y. Chen, Y. Li, H. Zhu, M. Gao, H. Hou, and C. Wang [5]

Y. Xin and co-authors published a comprehensive review in IEEE Access in 2018 on the use of machine learning and deep learning in cybersecurity. The paper surveys a wide range of techniques applied to intrusion detection, malware classification, and threat prediction. It highlights the strengths and limitations of various algorithms in dynamic and evolving threat environments.

H.J. Kim and J.S. Lim [6]

H.J. Kim and J.S. Lim presented a study in the IOP Conference Series: Materials Science and Engineering in March 2018. The research proposes a biometric authentication model based on ECG signals using a fuzzy neural network. This hybrid approach combines the interpretability of fuzzy logic with the learning capability of neural networks. The model aims to improve recognition accuracy and adaptability to signal variations.

J.R. Pinto, J.S. Cardoso, A. Lourenço, and C. Carrier [7]

J.R. Pinto, J.S. Cardoso, A. Lourenço, and C. Carreira published a study in Sensors in 2017 on ECG-based continuous biometrics. The research explores using ECG signals acquired from sensors embedded in a steering wheel for driver authentication. This approach enables unobtrusive and ongoing identity verification while driving. The study addresses signal quality, usability, and system feasibility in real-world conditions.

M. Sansone, R. Fusco, A. Pepino, and C. Sansone [8]

M. Sansone, R. Fusco, A. Pepino, and C. Sansone published a review in Journal of Healthcare Engineering in June 2013. The paper examines ECG pattern recognition techniques using artificial neural networks (ANNs) and support vector machines (SVMs). It compares the strengths and limitations of these methods in ECG classification and analysis.

CHAPTER 3: EXISTING SYSTEM

In the existing system to effectively determine the boundaries of datasets for electrocardiogram (ECG)-based biometric authentication, use case analysis is employed. By examining a range of practical application scenarios, three distinct use cases — or authentication categories — have been developed. These categories help define how the authentication system should behave under different conditions, whether it be continuous authentication, one-time login, or emergency identification. With the integration of more qualified and context-specific training data aligned to each use case, machine learning (ML) models can be trained more accurately, resulting in improved precision and reliability in ECG-based biometric authentication mechanisms.

Biometric authentication has garnered significant attention in recent years due to increasing demands for security and user convenience. Traditional schemes like fingerprint scanning and facial recognition have dominated the market, but they come with limitations such as spoofing risks and dependency on external physical features. In contrast, ECG-based authentication offers a unique advantage: it uses internal, live body signals that are inherently harder to forge or replicate. The dynamic nature of ECG signals not only enhances security but also allows real-time monitoring of a user's physiological state, making it a promising alternative for next-generation biometric systems.

Despite its advantages, ECG biometrics still face several critical challenges that need to be addressed. Key areas for further research include the categorization of authentication scenarios, the development of robust pre-processing techniques to enhance data quality, and the optimization of data acquisition methods to ensure consistency across different devices and environments. Additionally, careful selection of appropriate machine learning or deep learning classification approaches is essential to balance computational efficiency with classification accuracy.

CHAPTER 4: PROPOSED SYSTEM

This project introduces a comprehensive machine learning (ML) framework for ECG-based biometric authentication, aimed at addressing key challenges traditionally associated with ECG authentication methods. Recognizing the complexity of applying ECG authentication in real-world environments, the framework adopts a structured approach to mitigate issues such as inconsistent data quality, ineffective categorization of authentication scenarios, and selection of optimal machine learning techniques. By developing an integrated system that addresses these factors, the proposed framework strengthens the overall robustness, accuracy, and applicability of ECG-based biometric systems.

To better understand potential environments where ECG authentication can be deployed, the project emphasizes the importance of identifying basic application scenarios through systematic use case analysis. In the proposed framework, application environments for ECG authentication are classified into three general use cases: Hospital (HOS), Security Check (SCK), and Wearable Devices (WD). This categorization ensures that the framework can be tailored to suit the specific needs and operational characteristics of different sectors — whether it is patient monitoring in hospitals, identity verification at security checkpoints, or continuous authentication through personal wearable technology.

In preparation for effective model training, the framework introduces novel data pre-processing techniques, most notably the time slicing method. Time slicing divides continuous ECG signals into smaller, manageable segments, enhancing the consistency and quality of the training datasets for machine learning models. Alongside this, four new metrics have been developed for evaluating authentication precision and data quality: Mean Absolute Error Rate (MAER), Upper/Lower Range Control Limits (UCL/LCL), Accuracy Percentage within Range (APR), and Accuracy per Unit (APU). These metrics allow for a more detailed assessment of model performance and data

reliability, providing better insights during system development and evaluation.

An overview of the new framework reveals that it strategically integrates multiple machine learning techniques to optimize both regression and classification tasks within the core authentication process. For regression approaches, models such as Decision Trees (DT) and Support Vector Machines (SVM) are utilized to predict continuous-valued outputs related to ECG signal characteristics. For classification tasks, Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) are adopted to effectively differentiate between authorized and unauthorized users. The integration of the time slicing technique into the core process enhances the model's ability to learn temporal patterns in ECG signals, thereby further improving authentication precision and system reliability.

CHAPTER 5: MODULE IMPLEMENTATION

5.1 DATA FLOW DIAGRAM

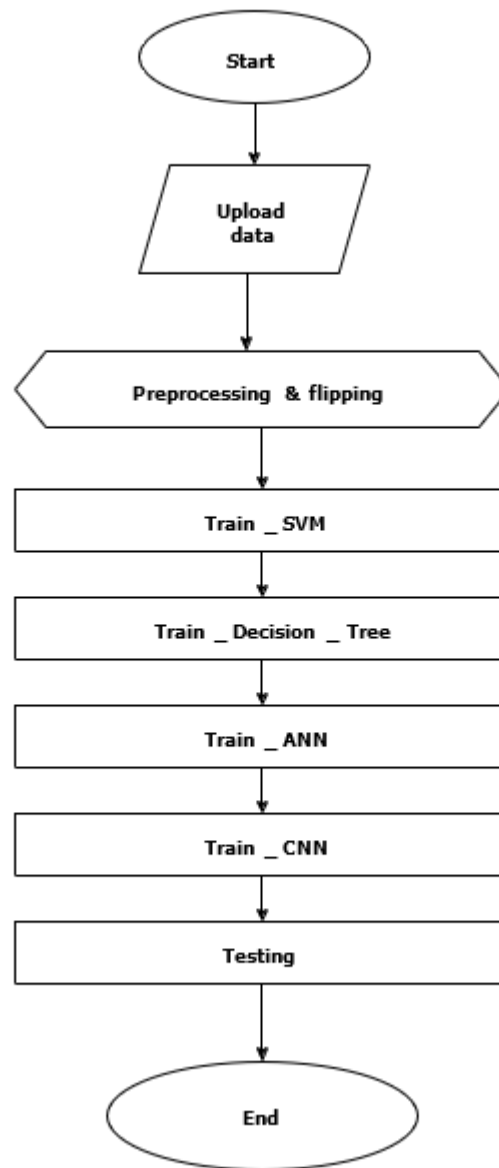


Fig 1:DATA FLOW DIAGRAM

This flowchart shows the steps of a machine learning project starting from the beginning. It begins with the "Start" box and moves to "Upload Data," where the user provides the data needed for training. Then it goes to "Preprocessing & Flipping," which means the data is cleaned and changed a bit (like flipping images) to help the models learn better. This step makes sure the data is ready to use.

After the data is prepared, it is used to train different types of models. First is the SVM (Support Vector Machine), which is good at telling things apart. Next is the Decision Tree, which makes decisions based on simple rules. Then comes the ANN (Artificial Neural Network), which can find more complex patterns. Lastly, the CNN (Convolutional Neural Network) is trained, which is good at understanding images.

In the final part, all the models are tested to see how well they perform. This happens in the "Testing" step. The results from each model may be compared or combined to get the best outcome. The process ends at the "End" box. This flowchart shows a full step-by-step way to try different models and choose the best one for making predictions.

5.2 CLASS DIAGRAM

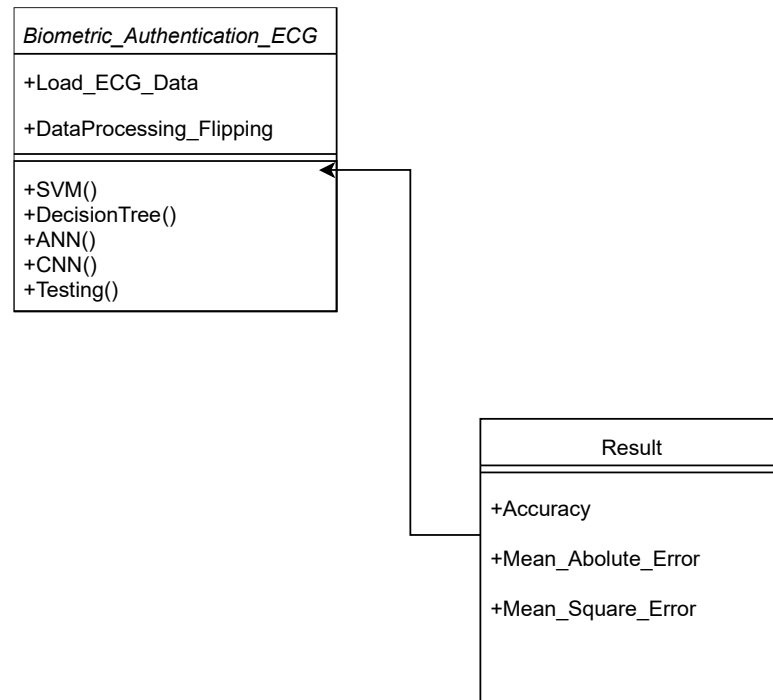


Fig 2:CLASS DIAGRAM

This diagram shows a system for biometric authentication using ECG (heart signal) data. It includes steps to load ECG data and prepare it by cleaning or modifying it (like flipping the signals) before training. These steps help the machine learning models learn better from the data.

In this system, different models are used to learn from the ECG data. These include SVM (Support Vector Machine), Decision Tree, ANN (Artificial Neural Network), and CNN (Convolutional Neural Network). Each model has its own method for training. There's also a **Testing()** step that checks how well each model performs using new, unseen data.

Another part of the system is the Result section, which stores the performance results. It includes things like Accuracy, Mean Absolute Error, and Mean Squared Error. These help in comparing the models to see which one works best. This setup keeps the training and result-checking parts of the system organized and easy to understand.

5.3 SEQUENCE DIAGRAM

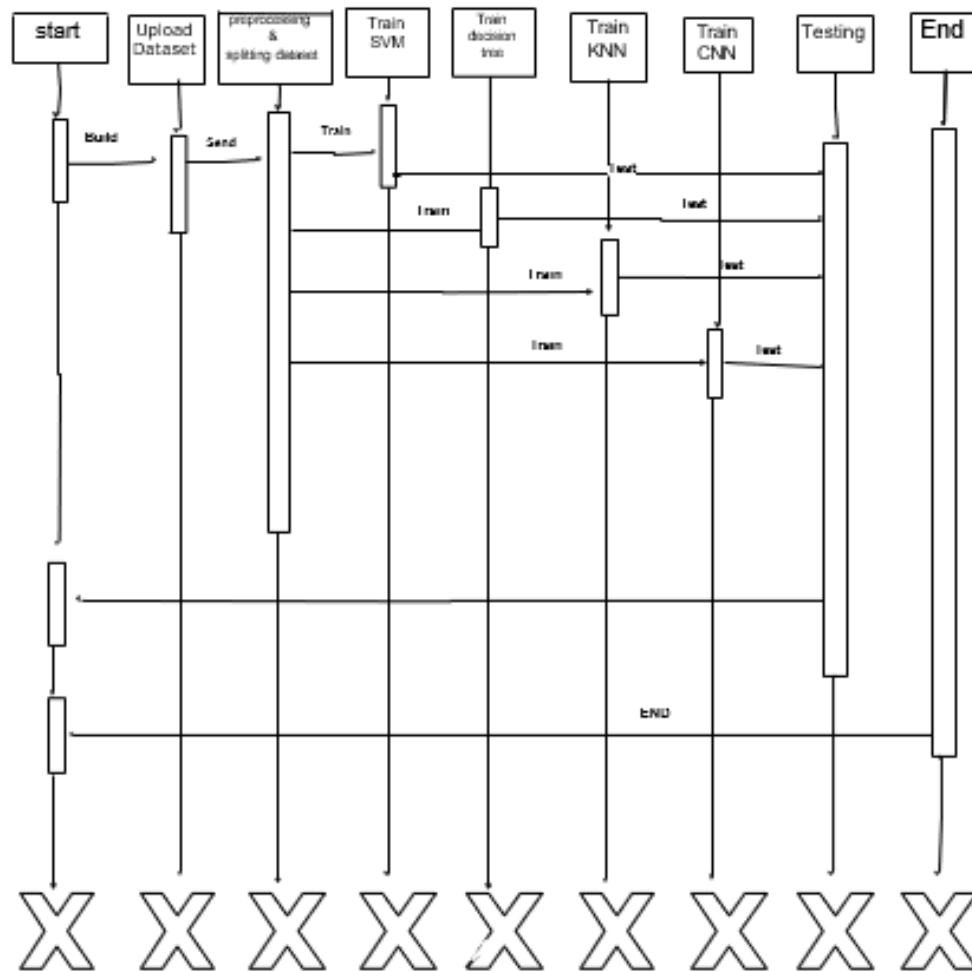


Fig 3: SEQUENCE DIAGRAM

This diagram is a sequence diagram that shows the step-by-step process of a machine learning system for ECG-based biometric authentication. It begins with the Start stage, where the system gets ready, and moves on to Upload Dataset, where ECG data is uploaded. Then, the data goes through Preprocessing & Flipping, where it's cleaned and prepared. This step ensures the data is suitable for training by improving its quality and increasing its size using techniques like flipping.

Next, the pre-processed data is used to train four different machine learning models: SVM, Decision Tree, ANN, and CNN. The diagram shows the flow of data between these stages—each model receives the training data, processes it, and then passes the test results forward. This setup allows the

system to compare the performance of different models. Arrows represent actions like training, testing, and passing data between components.

In the final part, the system performs Testing using the trained models. The results of these tests are analysed, and then the process reaches the End stage. Each vertical line in the diagram represents the "life" of each step or model, and the X at the bottom means that step is complete. This clear, step-by-step diagram helps visualize how ECG data moves through the system, how models are trained and tested, and how the entire process is completed from start to finish.

5.4 USE CASE DIAGRAM

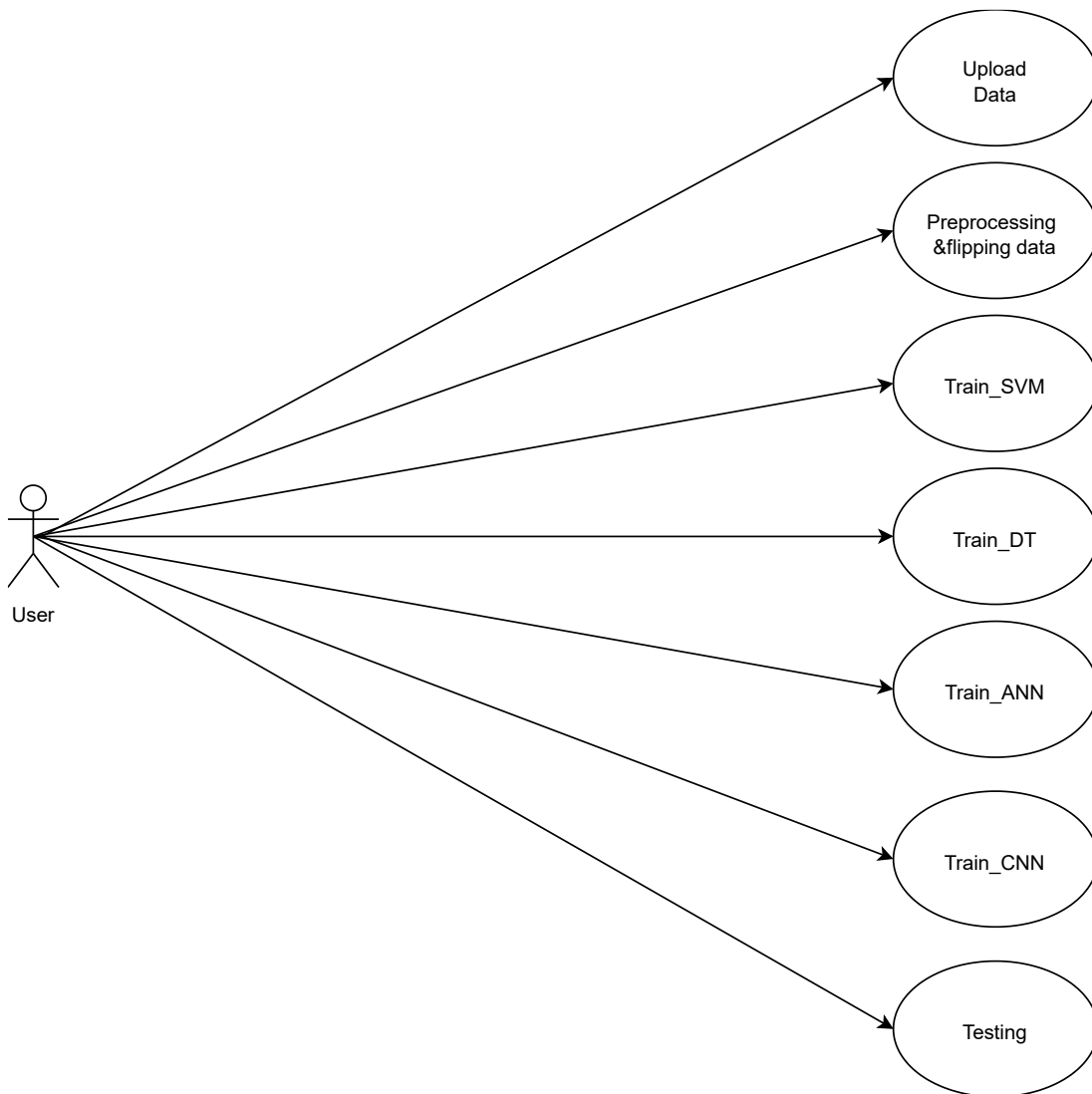


Fig 4: USE CASE DIAGRAM

This diagram is a use case diagram that shows the interaction between a User and a system for ECG-based biometric authentication. The stick figure on the left represents the user, who performs various actions within the system. Each oval shape represents a task or function that the user can trigger or be involved in. This kind of diagram is often used in software design to show what a user can do with a system.

The user starts by uploading the dataset, which is the first step to provide ECG data to the system. Then the user initiates preprocessing and flipping of the data, which involves preparing the ECG data by cleaning and possibly

flipping it to improve training. These steps are important to make sure the data is ready and in good shape before being used by the machine learning models.

After preparing the data, the user can train four different models: SVM (Support Vector Machine), DT (Decision Tree), ANN (Artificial Neural Network), and CNN (Convolutional Neural Network). The user also performs the Testing step, where the trained models are checked to see how well they work. Overall, the diagram shows that the user is actively involved in all the main steps of the system—from data preparation to training and testing models. It helps developers understand what the user expects to do when using the system

5.5 ACTIVITY DIAGRAM

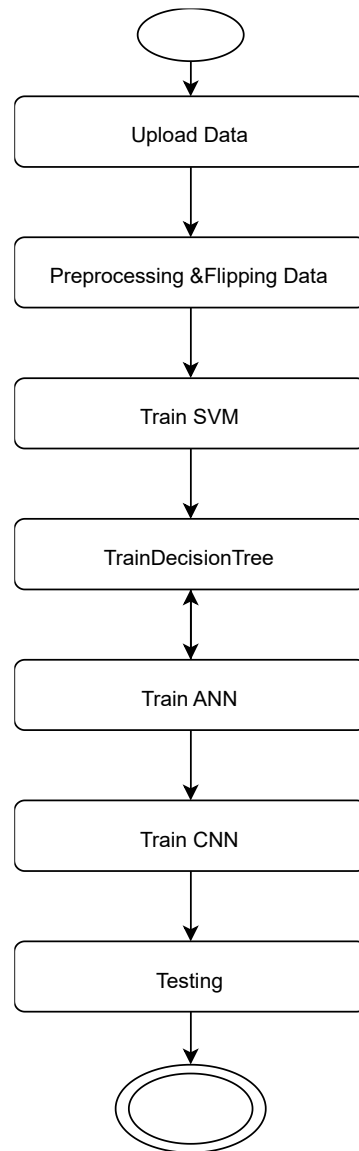


Fig 5: ACTIVITY DIAGRAM

The diagram shown is a flowchart that outlines the step-by-step process of a machine learning pipeline for ECG-based biometric authentication. It begins with a solid black circle, which marks the starting point of the process. The first action is “Upload Dataset,” where the user or system loads the ECG data required for training machine learning models.

After the data is uploaded, the next step is “Preprocessing and Flipping Data.” This involves preparing the ECG data by cleaning it, handling missing values, normalizing it, and possibly applying augmentation techniques like

flipping signals. This step is important because it improves the quality and variety of the data, which helps the machine learning models learn more effectively.

The flow continues through four training stages: first SVM (Support Vector Machine), then Decision Tree, followed by ANN (Artificial Neural Network), and finally CNN (Convolutional Neural Network). Each model is trained one after the other using the prepared data. Once all models have been trained, the system performs “Testing” to evaluate their performance on new, unseen data. The process ends at the black circle at the bottom, which represents the completion of the workflow. This clear, linear flow makes it easy to understand how data moves through each stage of the sytem.

CHAPTER 6: SYSTEM REQUIREMENTS

SOFTWARE AND HARDWARE REQUIREMENT SPECIFICATION

6.1 HARDWARE REQUIREMENTS

- Processor Type: Pentium -IV 2.4 GHz
- RAM: 512 MB
- Hard disk: 40 GB
- Floppy Drive: 1.44 MB
- Monitor: 15 VGA Colour

6.2 SOFTWARE REQUIREMENTS

- Operating System: Windows
- Coding Language: Python 3.7
- Tools Used: Python

CHAPTER 7: RESULT AND ANALYSIS

In this section, we present the results of applying our machine learning framework to electrocardiogram (ECG) data for biometric authentication. The objective was to assess the system's ability to identify or verify individuals based on their unique heart signal patterns. We evaluated the framework using a collected ECG dataset and measured its performance using key metrics such as accuracy, precision, recall, and F1-score. These metrics provide insights into how effectively the system can authenticate users and the frequency of errors.

Furthermore, we analyzed the performance of various machine learning models trained on the ECG data. By comparing the results across different models, we identified the most effective approach for this type of biometric task. This comparative analysis enabled us to understand the strengths and limitations of each model in the context of ECG-based authentication.

The section includes visual representations such as graphs and tables, along with detailed discussions that elucidate the system's performance. These analyses highlight both the strengths and weaknesses of the system, providing a comprehensive understanding of its capabilities and areas for improvement.

By examining the outcomes, we can pinpoint the advantages and drawbacks of each approach. This understanding serves as a foundation for enhancing the system and guiding future research aimed at making ECG-based biometric authentication more accurate and reliable.

In conclusion, the insights gained from this analysis can help improve future designs and highlight areas where the system can be further optimized. By leveraging the strengths of various machine learning models and addressing their limitations, we can develop more robust and efficient ECG-based biometric authentication systems.

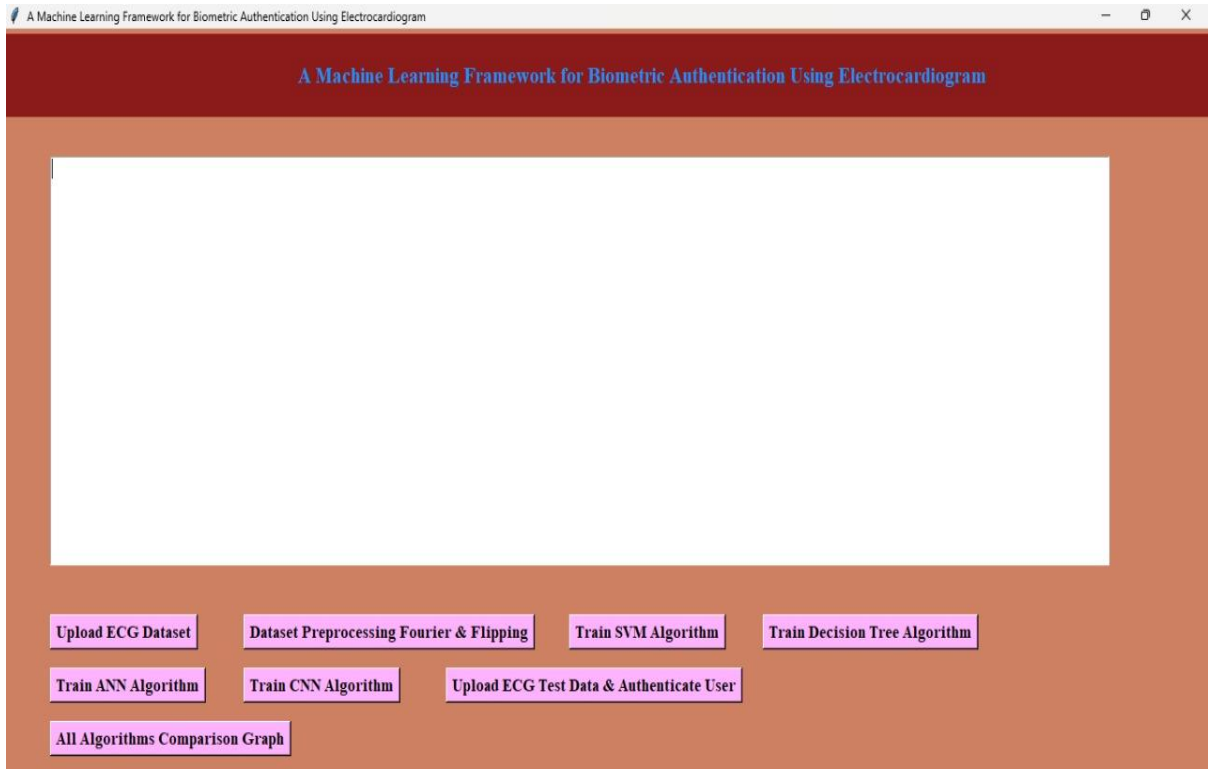


Fig 6: GUI of ECG-Based Biometric Authentication Using Machine Learning

The above image shows a simple software screen for a machine learning system that uses ECG (heart signal) data to check a person's identity. It has buttons to upload data, prepare it, train different types of models (like SVM, ANN, CNN, and Decision Tree), and compare how well they work. The big white box in the centre probably shows messages or results.

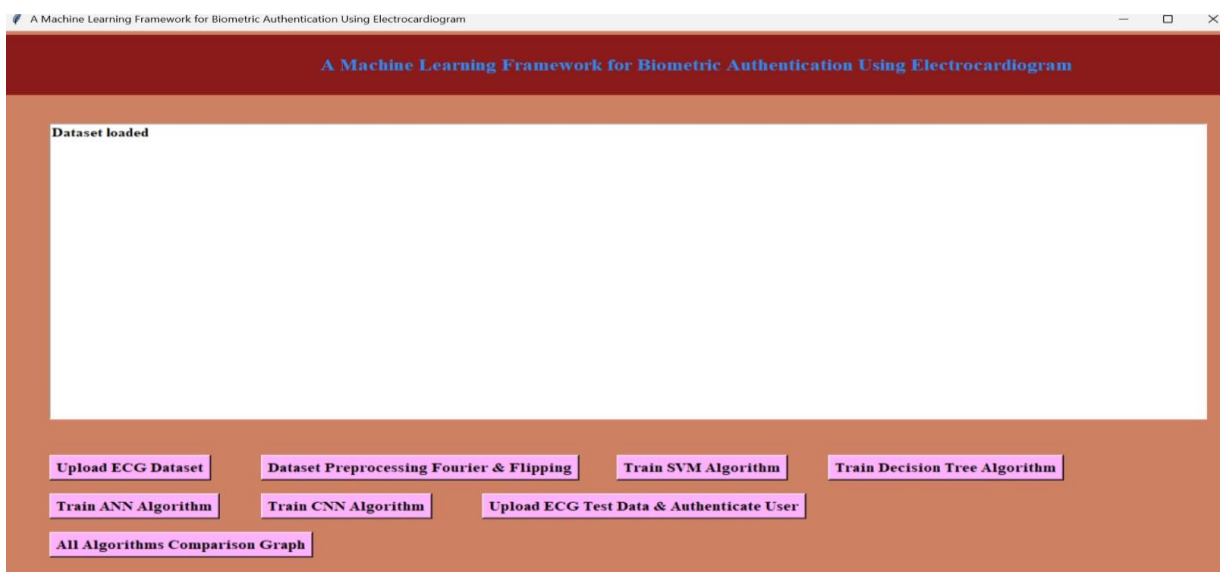


Fig 7: ECG Dataset loaded in Authentication GUI

The image displays a user interface for a machine learning system designed for biometric authentication using ECG data. It shows that the ECG dataset has been successfully loaded into the system. Users can now proceed to the next steps.

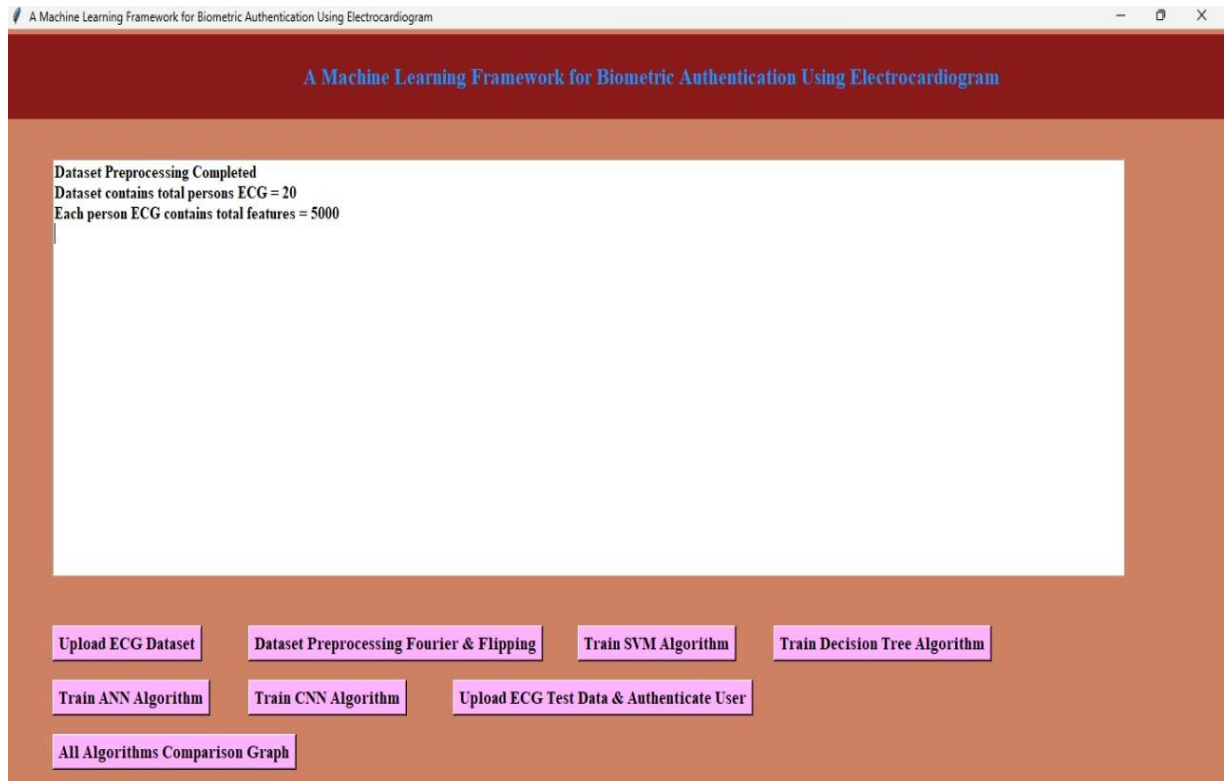


Fig 8: Pre-processed ECG Dataset with Feature Count

The image shows a software interface for biometric authentication using ECG signals. It indicates that the dataset preprocessing is complete, containing ECG data for 20 individuals, with each person's ECG having 5000 features.

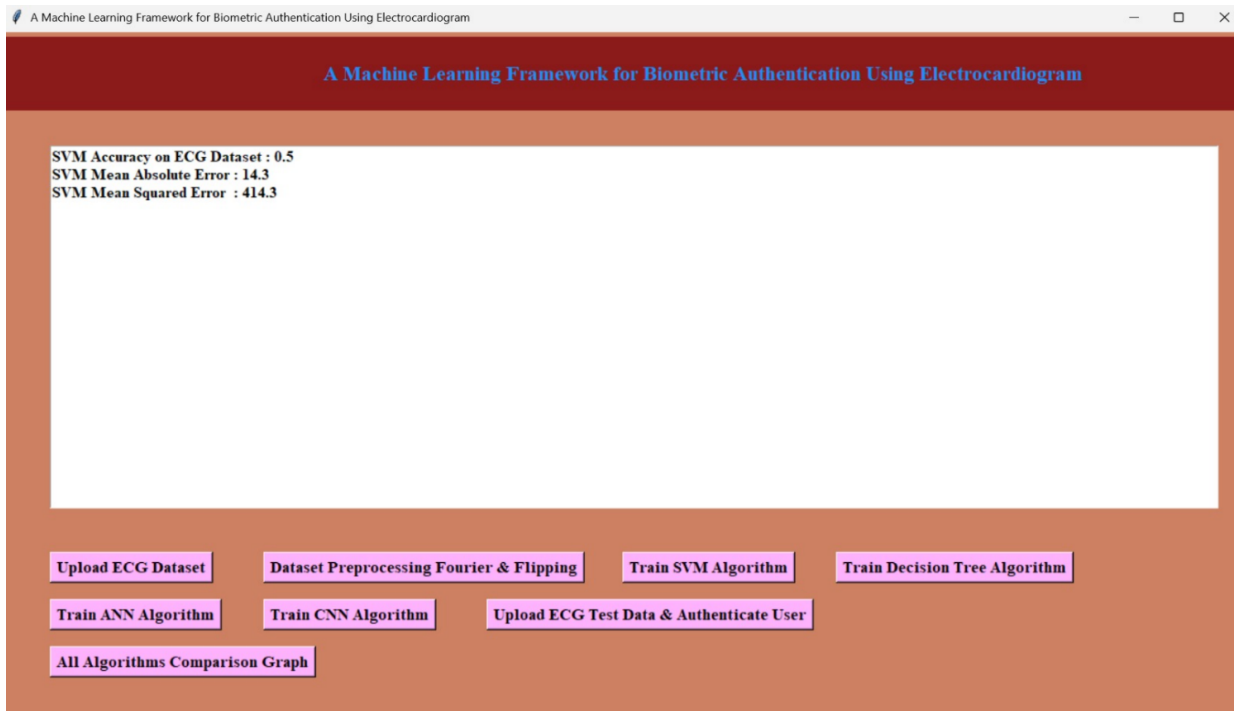


Fig 9: SVM Evaluation Result

The image shows the results after training a Support Vector Machine (SVM) algorithm on the ECG dataset. The reported performance metrics for the SVM on the ECG data are an accuracy of 0.5, a Mean Absolute Error of 17.0, and a Mean Squared Error of 591.6. This suggests the SVM's current performance on this specific task might need further improvement.

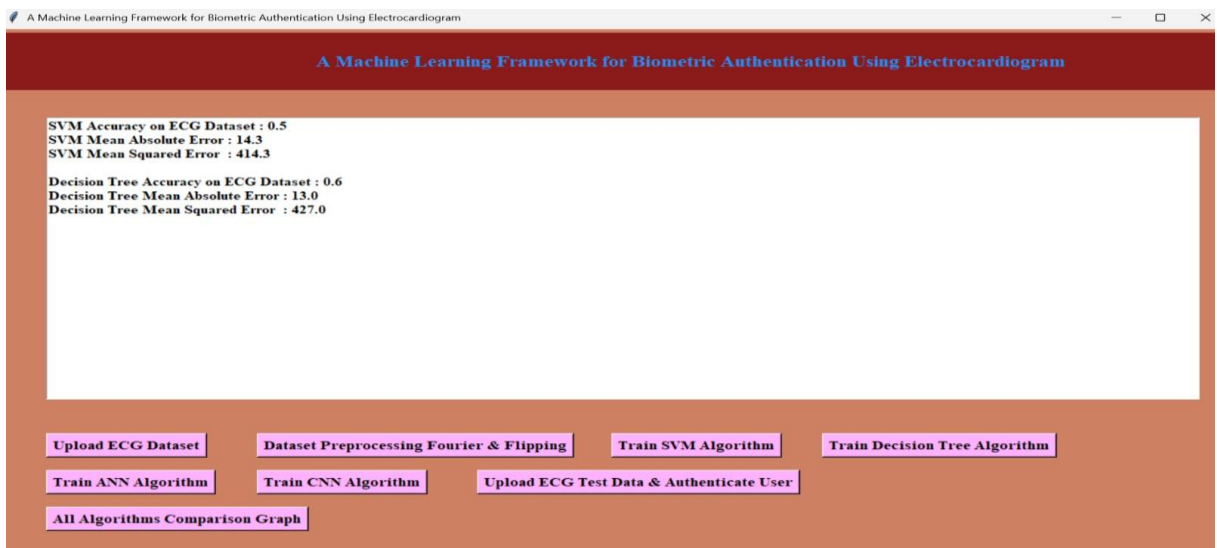


Fig 10: Evaluation of SVM and Decision Tree Models

The image now includes performance metrics for a Decision Tree algorithm trained on the same ECG dataset. The Decision Tree achieved an accuracy of 0.6, a Mean Absolute Error of 13.6, and a Mean Squared Error of 464.4. Comparing it to the SVM results, the Decision Tree shows a slightly better accuracy and lower error values on this data.

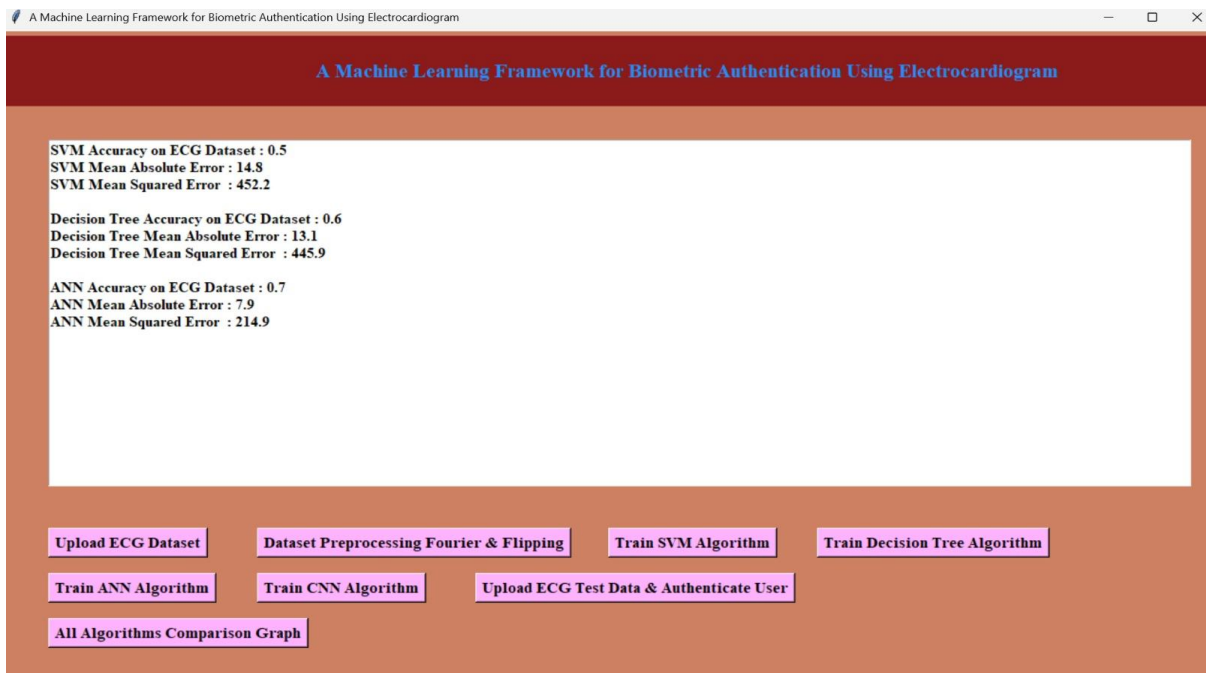


Fig 11: Evaluation report of SVM, Decision Tree and ANN

The image displays the performance of an Artificial Neural Network (ANN) algorithm on the ECG dataset. The ANN achieved an accuracy of 0.7, with a Mean Absolute Error of 7.9 and a Mean Squared Error of 214.9. Among the three algorithms shown so far (SVM, Decision Tree, and ANN), the ANN demonstrates the highest accuracy and the lowest error values on this ECG biometric authentication task.

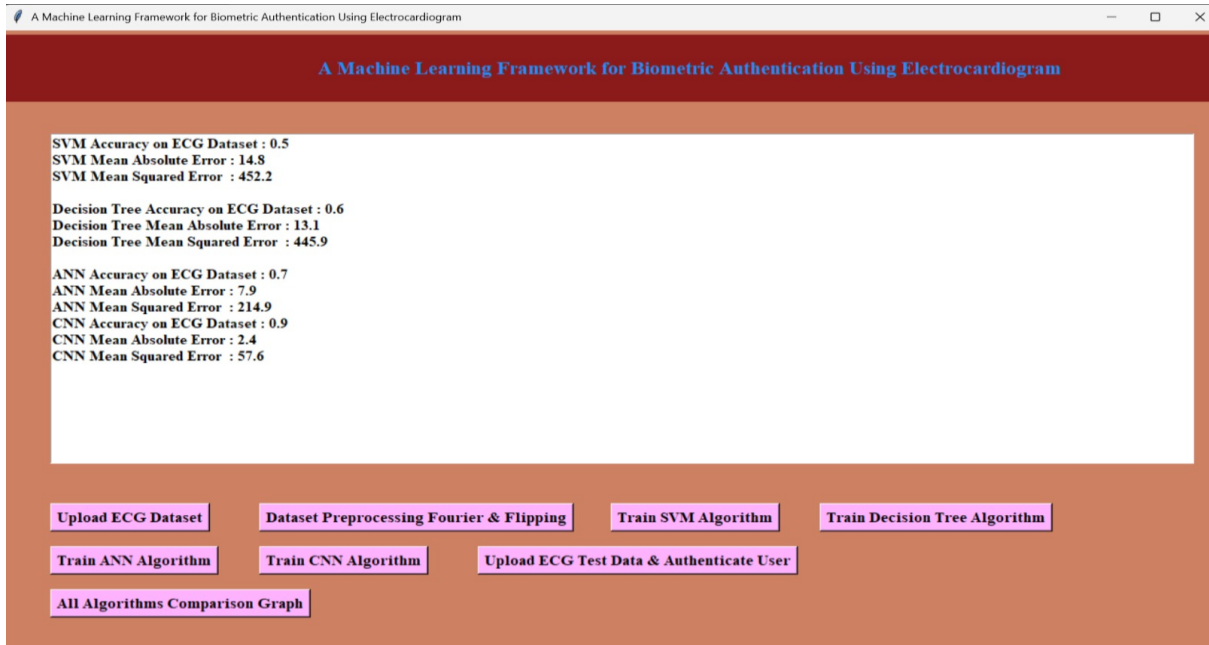


Fig 12: Comparative Analysis of ML Models Using ECG Data

The image now shows the performance of a Convolutional Neural Network (CNN) algorithm on the ECG dataset. The CNN achieved the highest accuracy so far at 0.9, with the lowest Mean Absolute Error of 2.4 and Mean Squared Error of 57.6. This indicates that the CNN is performing the best among the tested algorithms for biometric authentication using this ECG data.



Fig 13: User Interface for ECG Dataset processing and Model Training

The image shows the result of an ECG-based biometric authentication process. An ECG was uploaded and authenticated by the system. The system determined that the uploaded ECG belongs to Person ID: 0.

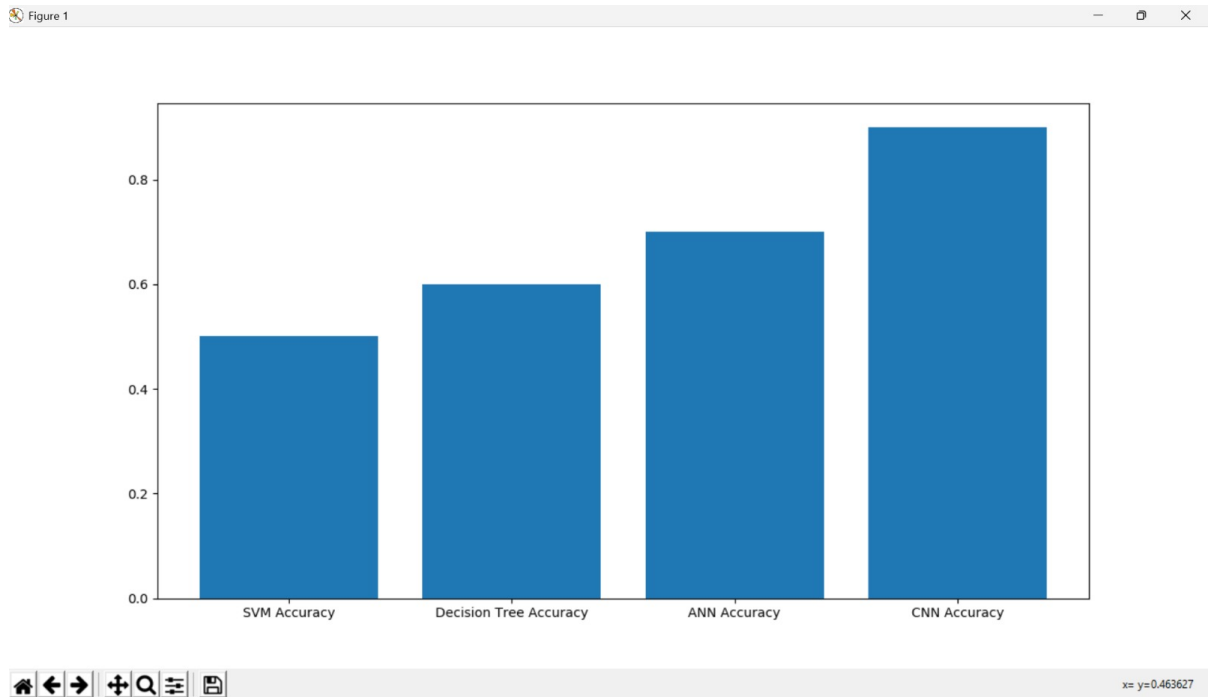


Fig 14: Performance Analysis of Machine Learning Models

The image is a bar graph comparing the accuracy of four different machine learning algorithms. The algorithms are Support Vector Machine (SVM), Decision Tree, Artificial Neural Network (ANN), and Convolutional Neural Network (CNN). The graph visually shows that the CNN algorithm achieved the highest accuracy and SVM achieved the least accuracy among the four.

CHAPTER 8: CONCLUSION & FUTURESCOPE

8.1. CONCLUSION

As new ECG detection devices become portable, lightweight, embeddable with smartphones and wearable devices, and connectable with remote servers through wireless technologies in the near future, ECG based biometric authentication will be deployed on massive application systems all over the world. To get high accuracy on user authentication, ML techniques are generally adopted to build a more robust evaluation model for ECG based biometric authentication. In this project a generalized machine learning framework for ECG based biometric authentication is introduced. The proposed framework describes the general data processing flow of a ML-based ECG authentication mechanism along with various function features to help researchers easily design and evaluate a ML- based ECG user authentication scheme. Those functions include three general authentication categories for ECG user authentication, three new data pre-processing techniques, and a time slicing technique to generate high quality ECG datasets. For people using ML technologies to investigate other topics instead of ECG based biometric authentication, several data pre-processing techniques and newly defined measure metrics offered by the proposed framework are still useful and can help researchers accelerate the development of their ML-based schemes.

8.2 FUTURE SCOPE

The future of ECG-based biometric authentication using machine learning lies in advancing adaptive algorithms that handle physiological variability, deploying privacy-preserving techniques for secure processing, and achieving regulatory compliance for sensitive biometric data. Continued research aims to enhance signal quality, optimize real-time model performance for edge devices, and integrate multimodal biometric systems to support scalable, seamless, and secure identity verification across digital platforms.

1.Improved Model Robustness and Generalization: ECG signals vary due to physical activity, emotional state, electrode placement, and health conditions.

2.Real-Time and Lightweight Models for Edge Devices: High computational cost of deep models limits deployment on wearables.

3.Multi-Modal Biometrics Integration: Relying solely on ECG may not meet high-security standards.

4.Cancelable and Privacy-Preserving Biometric Techniques: ECG is a permanent biometric (data can't be reset).

5.Better Accuracy in Real-Life Conditions

ECG signals can change due to emotions, movement, or health. Future systems will use smarter algorithms to recognize people accurately even when their heart signals change.

6.Works Well on Any Device

ECG readings may differ between devices. Future models will be designed to work reliably across different smartwatches, fitness bands, or medical devices.

7.Faster and Lighter Models for Small Devices

Deep learning models are heavy. In the future, we can make lightweight models that run quickly on small devices like smartwatches or phones without needing the internet.

8.Use with Other Biometrics

Relying only on ECG might not always be enough. In the future, ECG will be combined with other biometrics like face, voice, or fingerprint for better security.

9.Safe and Private Use of ECG Data

Since ECG data can't be changed like a password, future systems will use special techniques to protect your heart data and keep it private and secure.

10.Adapts as Your Body Changes

Your heart signal may change as you age or if you get sick. Future systems will learn and adjust over time to keep working accurately.

11.Detects Health Issues Too

Besides identifying you, future ECG systems might also alert you if they detect heart problems like irregular heartbeat or stress.

12.Easy to Use Without Manual Setup

Future systems will work automatically without needing you to set them up again every time or wear the device in a specific way.

13.Approved for Use in Healthcare and Finance

In the future, ECG authentication systems could be officially approved for use in hospitals, banks, and secure government services.

14.Works Across the World with Common Standards

Researchers and companies will work together to create shared rules and formats so ECG systems can work smoothly on any device, anywhere in the world.

CHAPTER 9 : APPENDIX

9.1. TECHNOLOGIES USED : INTRODUCTION TO PYTHON

Python is a widely used high-level programming language, created by Guido Van Rossum and first released in 1991. An interpreted language, Python has a design philosophy that emphasizes code readability and syntax that allows the programmer to express concepts in fewer lines of code that might be used in programming languages such as C++ or Java. It provides and constructs models that enable clear programming on both small and large scales. Python features a dynamic type system & automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library. Python interpreters are available for many operating systems. The reference implementation of Python is open source software and has a community-based development model, as do nearly all of its variant implementations. Programming languages have been for ages, and every decade sees the launch of a new language sweeping the developers off their feet. Below are the major features and applications due to which people choose Python as their first programming language:

1. Python's popularity and high salary
2. Python is used in Data Science
3. Python's scripting and automation
4. Python used with Big Data
5. Python supports Testing
6. Computer Graphics in Python
7. Used in Artificial Intelligence
8. Python in Web Development
9. Python is portable and extensible

Features of python

1. **Easy-to-learn:** Python has few keywords, simple structure, and a clearly defined syntax. It allows the student to pick up the language quickly.
2. **Easy-to-read:** Python code is more clearly defined and visible to the eyes.
3. **Easy- to-maintain:** Python's source code is fairly easy-to-maintain.
4. **.A broad standard library:** Python's bulk library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh
- 5.**Interactive Mode:** Python has support for an interactive mode which allows the interactive testing and debugging of code snippets.
6. **Portable:** Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- 7.**Extendable:** We can add low-level modules to the Python interpreter. These modules enable the programmers to add or customize their tools to be more efficient.
8. **Databases:** Python provides interfaces to all major commercial databases.
- 9.**GUI Programming:** Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X window system of UNIX.
- 10.**Scalable:** Python provides a better structure and support for large programs than shell scripting. It supports functional and structured programming methods as well as OOP.
11. It supports automatic garbage collection.
12. It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java

Machine Learning :

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *unable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data. Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and hasn't surpassed human intelligence in many aspects. Then the question is what is the need to make machine learning? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale". Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning: -

- **Quality of data** – Having good-quality data for ML algorithms is one of the biggest challenges.
- Use of low-quality data leads to the problems related to data preprocessing and feature extraction.
- **Time-Consuming task** – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and

- retrieval. **Lack of specialist persons** – As ML technology is still in its infancy stage, availability of expert resources is a tough job.
- **No clear objective for formulating business problems** – Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.
- **Issue of over fitting & under fitting** – If the model is overfitting or under fitting, it cannot be represented well for the problem.
- **Curse of dimensionality** – Another challenge ML model faces is too many features of data points. This can be a real hindrance
- **Difficulty in deployment** – Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning: -

Machine Learning is the most rapidly growing technology. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML –

1. Emotion analysis
2. Sentiment analysis
3. Error detection and prevention
4. Weather forecasting and prediction
5. Stock market analysis and forecasting
6. Speech synthesis
7. Speech recognition
8. Customer segmentation
9. Object recognition
10. Fraud detection

9.2. Time Slicing and Machine Learning

- The time slicing technique is considered for building up the dataset of the machine learning training. This approach is especially applicable for building

up the machine learning training data. The ECG data are sliced based on a slice (window) time (typically known as a sliding window) with the R- peak anchoring.

- This method could generate enough data samples and each sliced data becomes a sample input for the machine learning training. The time sliced ECG dataset is very flexible not only to mix with other training inputs but also to apply various ML training methods which is explained in the subsection.

Time Sliced ECG Data for Machine Learning

13. The QRS complex is the combination of three of the graphical deflections seen on a typical ECG which is usually the central and most visually obvious part of the tracing. An R-peak which is the maximum of a QRS complex. It indicates one heartbeat and the moment of the R-peak are commonly used for the anchor of the QRS complex including the R- peak detection and optimizing the sliding window time. Time slicing, which is basically slicing ECG data in the time domain, is targeted for chopping the ECG signal from an R-peak moment to the sliding window period and layering these Pieces based on the R- peak moment (i.e., R-peak anchoring). Each slicing piece based on the R-peak anchoring in figure becomes sample inputs for the machine learning training. In this project the average minimum of a heartbeat interval from a typical heart rate is chosen as a slicing time (i.e., sliding window) which is 0.6 seconds which is equivalent with 100 bps heartbeat rate. The optimized sliced window time depends on the purpose of ECG based projects

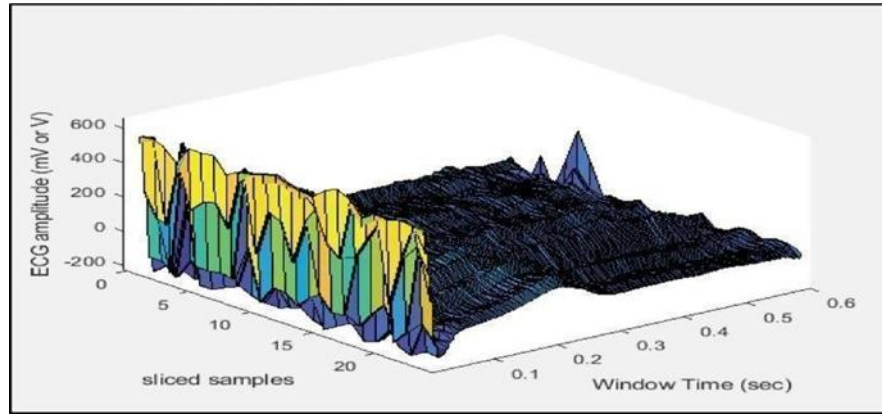


Fig 15: Time Sliced ECG data

9.3. Machine Learning Process for Training Data

Machine learning is a subset of artificial intelligence (AI) which computer systems use to effectively perform a specific task without using explicit instructions. Machine learning algorithms build a mathematical model of training data to make predictions or decisions without being explicitly programmed. It has been widely studied for analyzing ECG data by using AI techniques.

This research focuses to train the time sliced ECG data by using ML techniques. Various techniques are provided to demonstrate how a machine learning model is applied into the ECG based biometric authentication.

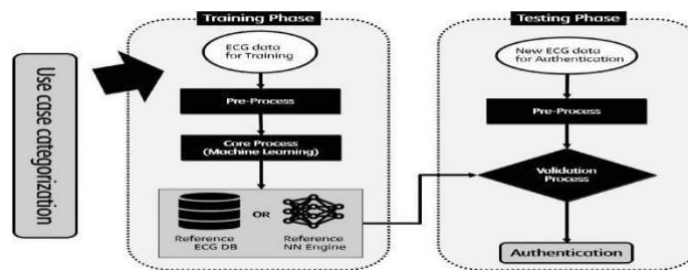


Fig 16: Training and Testing data

Machine Learning for Regression Approach

A multi-variable regression is a data mining task of predicting a value of the target by building a model based on multiple variables. The Support Vector Machine (SVM) is a discriminative classifier formally defined by a separating pattern. The Decision Tree (DT) could also build a regression model in the forms of a tree structure. It breaks down a dataset into smaller

subsets while at the same time an associated decision tree is incremented. It is shown that the DT technique delivers better performance compared to the SVM technique for building the regression modeling based on the dataset. Besides these two ML techniques, several other machine learning models could be applied for developing a regression model. In addition, choosing a DT model might be the best option only dedicated for a particular setup of computer systems and particular (time sliced) ECG datasets. The optimal selection of a proper ML model for a regression approach might vary by a different dataset or by a different computing system.

9.4. Machine Learning for Classification Approach

Classification is the target to identify the category set based on the basis of a training set of data containing observations. The set of the identification could be considered as a set of categories and each identification becomes each category set. Neural network (NN) models (either Artificial Neural Networks or Convolutional Neural Networks) could be designed by using the time sliced ECG dataset as the inputs. Regardless of which NN models are applied, both neural network models are compatible to use the same input nodes and the same output nodes. Therefore, the time sliced ECG dataset based on the R-peak anchoring could be the input samples for both neural network models. It is noted that this project does not deal with any particular models and the project is only targeted to provide the guidelines to design the better NN for authentication projects.

The following section explains the data quality measures to check the quality of both a machine learning models and training dataset.

9.5. Data Quality Measures

Some performance measures (evaluation metrics) are the values of evaluation results of sample data qualities and regression approaches-based authentication systems. Typical data quality measures and performance

measures for a regression approach are as follows:

- Sum of Squares Error
- Sum of Squares Total
- Mean Squared Error
- Mean Absolute Error

Besides typical quality measures, additional new measurement factors are introduced in the project. These quality measures are not only applied for evaluating regression approached machine learning systems but also applied for the criteria of validating samples. The new data quality measures are as follows:

- Mean Absolute Error Rate
- Upper and Lower Range Control Limits
- Accuracy Percentage within Ranges
- Accuracy per Upper Control Limit

The details of above new data quality measures are explained in the following subsections.

9.6. Mean Absolute Error Rate (MAER):

The **Mean Absolute Error Rate (MAER)** is a normalized regression metric that measures the relative error between predicted values and their moving mean references. MAER offers intuitive interpretability and helps detect data outliers. It supports both model performance evaluation and training data quality assessment.

$$\text{MAER} = 1/N \sum_{n=1}^N |Y_n - \mu_n| / \mu_n + \epsilon, \epsilon \sim$$

Accuracy Percentage within Ranges (APR):

The **Accuracy Percentage within Ranges (APR)** measures the proportion of ECG data samples that fall within an acceptable range (0 to UCL). It is calculated as the count of valid samples divided by the total number of samples. Higher APR values indicate better data quality and model performance.

$$\mathbf{APR=n(\{R\leq UCL\})/n(\Omega R)}$$

Accuracy per UCL (APU):

The Accuracy per UCL (APU) is the ratio of APR to the Upper Control Limit (UCL), indicating training improvement relative to control thresholds. It helps assess data and model quality even when UCL changes. APU serves as an alternative metric for evaluating regression-based ML systems, especially in biometric authentication.

$$\mathbf{APU=APR/UCL}$$

Advantages of Machine Learning: -

1. Easily identifies trends and patterns -Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans.
2. No human intervention needed (automation)-With ML you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn.

CHAPTER 10: BIBLIOGRAPHY

Zhang, Dzho, and X. Zeng Heart ID [1]

In 2017, Zhang, Dzho, and X. Zeng published a paper titled *Heart ID* in the journal *IEEE Access*. The study introduced a multiresolution convolutional neural network (CNN) designed for human identification using ECG (electrocardiogram) signals. This method uses heart data as a biometric to recognize individuals. It is especially useful for smart health applications where secure and accurate identification is important.

J.R. Pinto, J. S. Cardoso, and A. Lourenço [2]

The paper by J.R. Pinto, J.S. Cardoso, and A. Lourenço, published in *IEEE Access* in 2018, provides a comprehensive overview of ECG biometrics. It traces the evolution of the field, emphasizing the unique advantages of ECG signals for biometric recognition. The authors discuss key challenges such as intra-subject variability, noise, and acquisition constraints.

E.J. da Silva Luz, G.J.P. Moreira, L.S. Oliveira, W.R. Schwartz and D.Menotti [3]

E.J. da Silva Luz and colleagues published a study in *IEEE Transactions on Information Forensics and Security* in May 2018. The paper focuses on learning deep representations for heart biometrics using off-the-person ECG data. It emphasizes the use of deep learning techniques to extract robust features for identification.

H. Kim and S.Y. Chun [4]

H. Kim and S.Y. Chun presented a study in *IEEE Access* in 2019 on cancellable ECG biometrics. The work introduces a method combining compressive sensing with a generalized likelihood ratio test (GLRT). This

approach enhances both security and privacy by allowing biometric templates to be revocable.

Y. Xin, L. Kong, Z. Liu, Y. Chen, Y. Li, H. Zhu, M. Gao, H. Hou, and C. Wang [5]

Y. Xin and co-authors published a comprehensive review in IEEE Access in 2018 on the use of machine learning and deep learning in cybersecurity. The paper surveys a wide range of techniques applied to intrusion detection, malware classification, and threat prediction. It highlights the strengths and limitations of various algorithms in dynamic and evolving threat environments.

H.J. Kim and J.S. Lim [6]

H.J. Kim and J.S. Lim presented a study in the IOP Conference Series: Materials Science and Engineering in March 2018. The research proposes a biometric authentication model based on ECG signals using a fuzzy neural network. This hybrid approach combines the interpretability of fuzzy logic with the learning capability of neural networks. The model aims to improve recognition accuracy and adaptability to signal variations.

J.R. Pinto, J.S. Cardoso, A. Lourenço, and C. Carrier [7]

J.R. Pinto, J.S. Cardoso, A. Lourenço, and C. Carreira published a study in Sensors in 2017 on ECG-based continuous biometrics. The research explores using ECG signals acquired from sensors embedded in a steering wheel for driver authentication. This approach enables unobtrusive and ongoing identity verification while driving. The study addresses signal quality, usability, and system feasibility in real-world conditions.

Institute Name:

Malla Reddy College of Engineering for women

Title of the Innovation/Prototype:

A MACHINE LEARNING FRAMEWORK FOR BIOMETRIC AUTHENTICATION USING ELECTROCARDIOGRAM

Team Lead Name:

GHATTI VENKATA SIVANY

Team Lead Email:

sivanyghatti05@gmail.com

Team Lead Phone:

8106419243

Team Lead Gender:

Female

FY of Development:

2024-25

Developed as part of:

Academic Requirement/Study Project

Innovation Type:

Service

TRL LEVEL:

2

Theme:

Smart Education, Infrastructure, Defence & Security,

Define the problem and its relevance to today's market / society / industry need:

This project introduces a framework for how to appropriately adapt and adjust Machine Learning (ML) techniques used to construct Electrocardiogram (ECG)-based biometric authentication schemes. The proposed framework can help investigators and developers on ECG-based biometric authentication mechanisms define the boundaries of required datasets and get training data with good quality. Based on various application scenarios on ECG-based authentication, use cases are developed. For researchers adopting ML techniques to design new schemes in other research domains, the proposed framework is still useful for generating the ML-based training and testing datasets with good quality and utilizing new measure metrics.

Describe the Solution / Proposed / Developed:

We propose a Machine Learning framework for ECG based biometric authentication in order to mitigate identified challenges on ECG authentication. To better understand potential application environments for ECG authentication, it is necessary to identify basic application scenarios through use cases. In the proposed framework application scenarios using ECG authentication are categorized into three general use cases: Hospital (HOS), Security Check (SCK) and Wearable Devices (WD).

Explain the uniqueness and distinctive features of the (product / process / service) solution:

A MACHINE LEARNING FRAMEWORK FOR BIOMETRIC AUTHENTICATION USING ELECTROCARDIOGRAM Utilization of ECG for Biometric Authentication Incorporation of Machine Learning Models Non-invasive and User-Friendly Enhanced Security Integration with Multi-factor Authentication (MFA) Privacy and Data Security Robustness to Noise and Artifacts Potential for Continuous Health Monitoring

How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any:

Illustrates an overview of the new framework model for ML-based ECG biometric authentication. Within the core process portion, several ML techniques are adopted: Decision Tree (DT) and Support Vector Machine (SVM) for regression approach, and Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) for classification approach. In addition, time slicing technique for ECG data is developed and associated with the core process.

Is there any IP or Patentable Component associated with the Solution?:

No

Has the Solution Received any Innovation Grant/Seedfund Support?:

No

Are there any Recognitions (National/International) Obtained by the Solution?:

No

***Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?:**

No

Had the Solution Received any Pre-Incubation/Incubation Support?:

No

Video URL:

https://drive.google.com/file/d/1hCa93PtQrYxHexO_3W66GLFUEgQl8wE1/view?usp=sharing

Downloaded on: 05-05-2025

This report is electronically generated against Yukti - National Innovation Repository Portal.